Hands-on Lab: Quantum Computing for Ubiquitous and Pervasive Computing Systems

From theory to simulation, measurement, and application.

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We Know the Algorithms. Now How Do We Run Them

1. Module 1 · Foundations of Quantum Circuits
2. Quantum Sensing, Perception & Localization
3. Module 2 · Mapping Pervasive Problems to Quantum Models
4. Module 3 · Case Studies and Applications
5. Panel · Module 4: Challenges, NISQ, and Future Roadmap

Learning Objectives

Tools: Python | Qiskit | Qiskit Aer | Matplotlib

1. Build
Construct quantum circuits.

2. Simulate
Run local Aer backends.

3. Measure
Extract bitstring counts.

4. Visualize
Plot via Matplotlib.

5. Evaluate
Analyze state fidelity.

Prepare the Quantum Computing Environment

Installation, core software stack, and execution targets

Install the required libraries:
python -m pip install qiskit qiskit-aer matplotlib

Our Software Stack

Python	Programming language
Qiskit	Quantum circuit construction
Qiskit Aer	Local quantum circuit simulation
Matplotlib	Visualization of measurement results

Execution Options

- **Local laptop simulation**
- **Cloud quantum hardware, if available**

What We Will Build

Seven practical exercises spanning foundational circuits to algorithm simulation

1. Basic circuit using standard quantum gates
2. Superposition demo via Hadamard gate
3. Entangled Bell state simulation
4. Visualizing measurement probabilities & distribution counts
5. Grover's search algorithm implementation
6. Smart-building access point quantum search
7. Quantum noise simulation & comparison

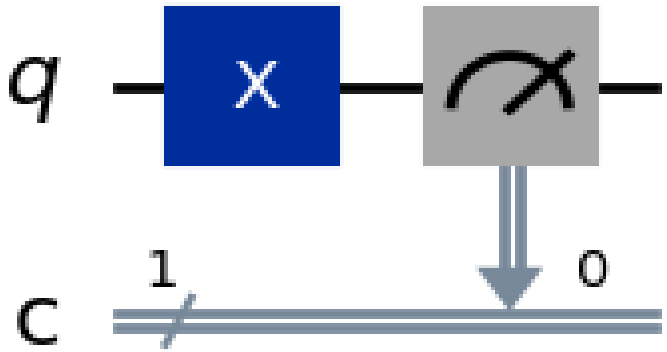
Practical 1: Create a Quantum Circuit Using Basic Gates

Objective: Construct and execute a one-qubit circuit

Circuit Operations

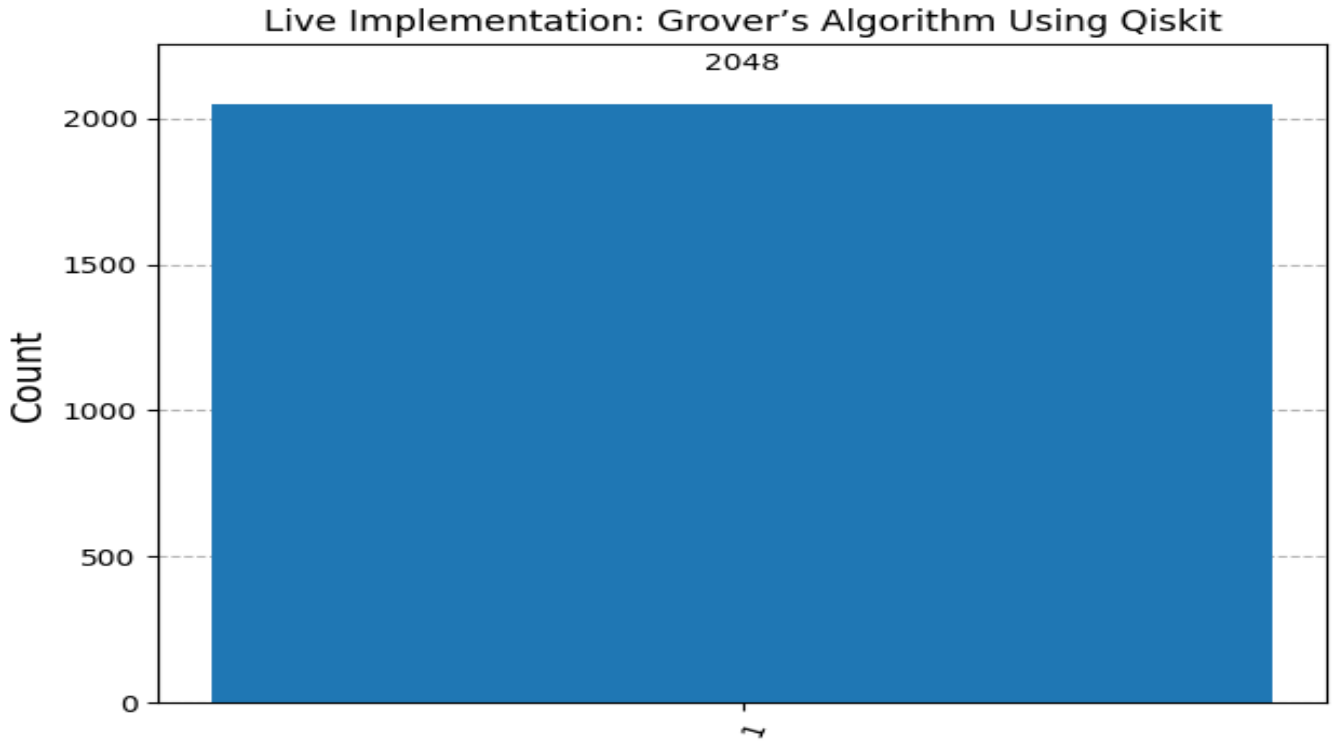
1. Initialize a qubit in the state |0⟩
2. Apply an X gate
3. Measure the qubit

Key Idea
Quantum gates transform a qubit's state, and measurement produces a classical result.



What the Code Demonstrates

Core quantum circuit operations, state transformations, and simulation workflow



Practical 2: Demonstrate Superposition

Objective: Explore the effect of a Hadamard gate

Circuit Flow & Transformation

|0⟩ → H → Measurement

Mathematical transformation

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Key Idea
Superposition produces a quantum state with multiple possible measurement outcomes.

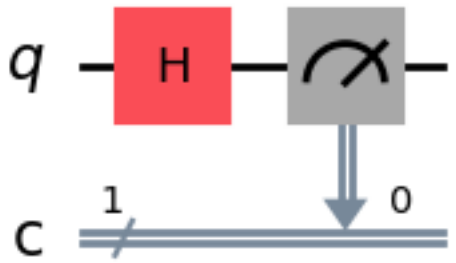
EXPECTED MEASUREMENT PROBABILITIES

50%	50%
P(0)	P(1)

Practical 2: Visualize Measurement Probabilities

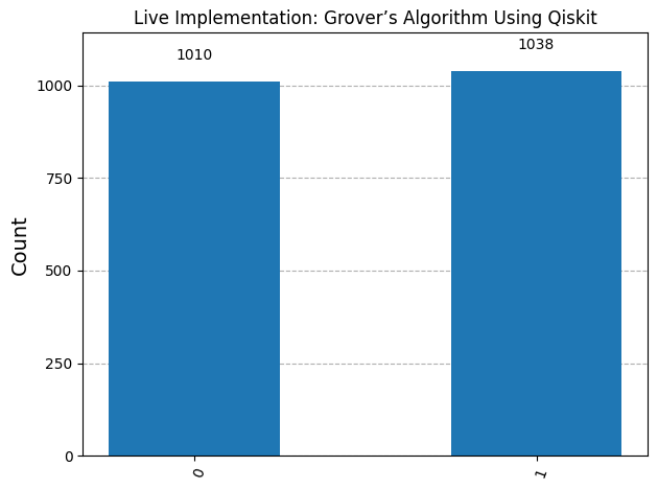
Objective: Verify the theoretical measurement probabilities using a local quantum simulator.

1. Quantum Circuit



Generated Qiskit circuit diagram

2. Measurement Histogram



Shot distribution plot (1024 shots)

3. Actual Output

```
# Jupyter Notebook Output
counts = {'00': 1010, '11': 1038}
shots = 2048
backend = 'aer_simulator'
```

Result: Exact 50 / 50 Correlation

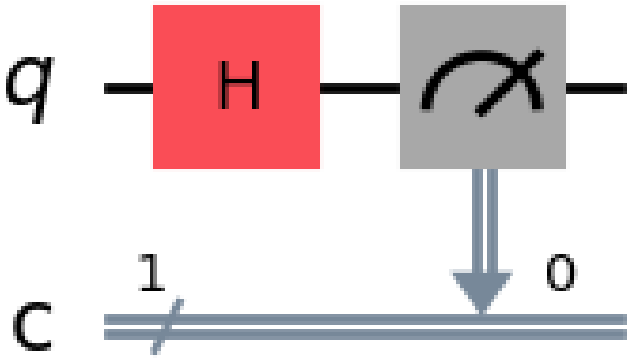
Practical 3: Build an Entangled Bell State

Objective: Create and measure a two-qubit entangled state

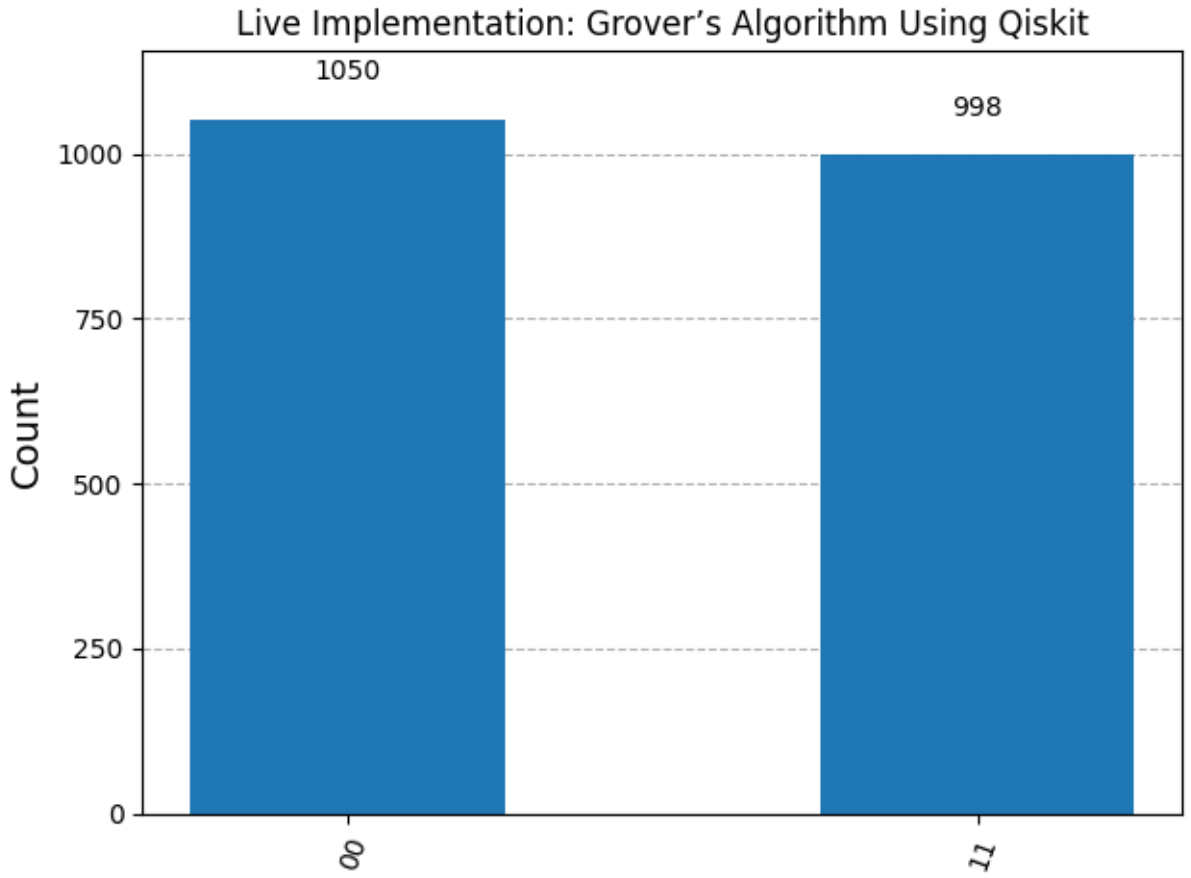
Circuit Operations & State

1. Apply H to qubit 0
2. Apply CX (control: q0, target: q1)
3. Measure both qubits

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



Key Idea
Entanglement creates deep correlations unexplainable by treating qubits independently.

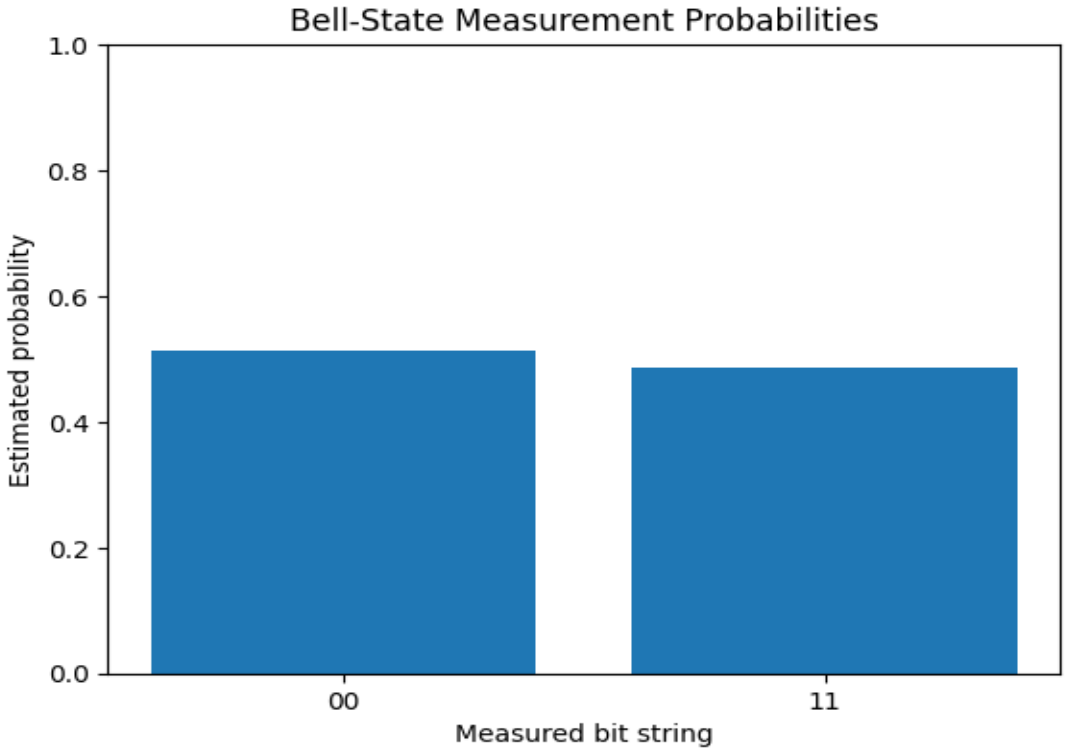

Live Implementation: Grover's Algorithm Using Qiskit
1050
998
1000
750
500
250
0
Count
00
11

Practical 4: Visualize Measurement Probabilities

Objective: Convert measurement counts into a visual representation

Procedure

1. Execute the circuit with repeated shots
2. Collect the measurement counts
3. Normalize the counts into probabilities
4. Plot the outcomes using Matplotlib



Practical 5: Search for a Target Using Grover's Algorithm

Objective: Implement a small quantum search algorithm across four items

Algorithm Workflow

1. Initialize two qubits
2. Create equal superposition of four states
3. Apply an oracle marking the target
4. Apply the Grover diffusion operator
5. Measure the qubits

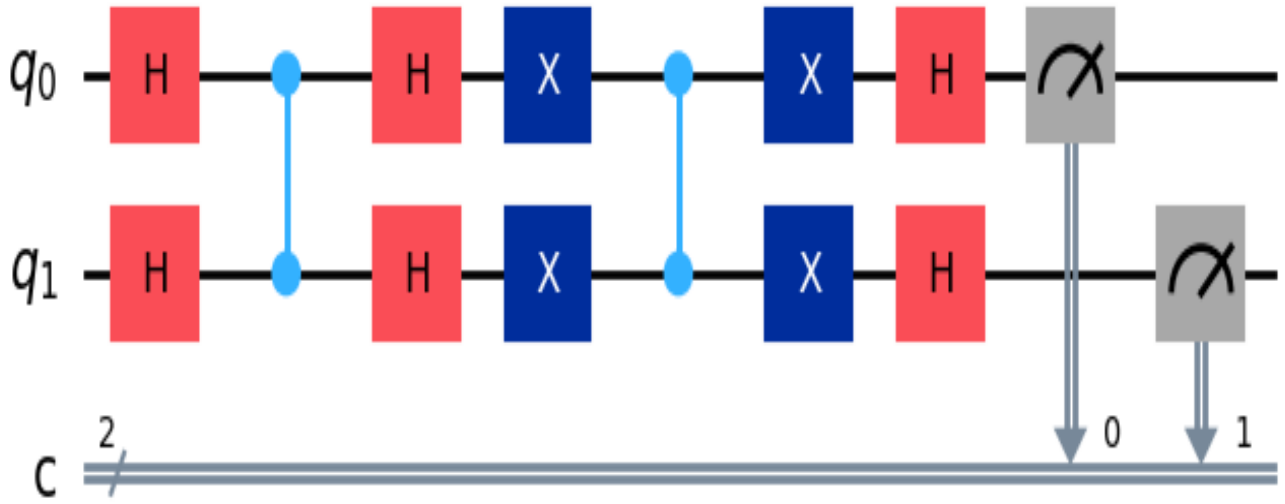
IDEAL RESULT

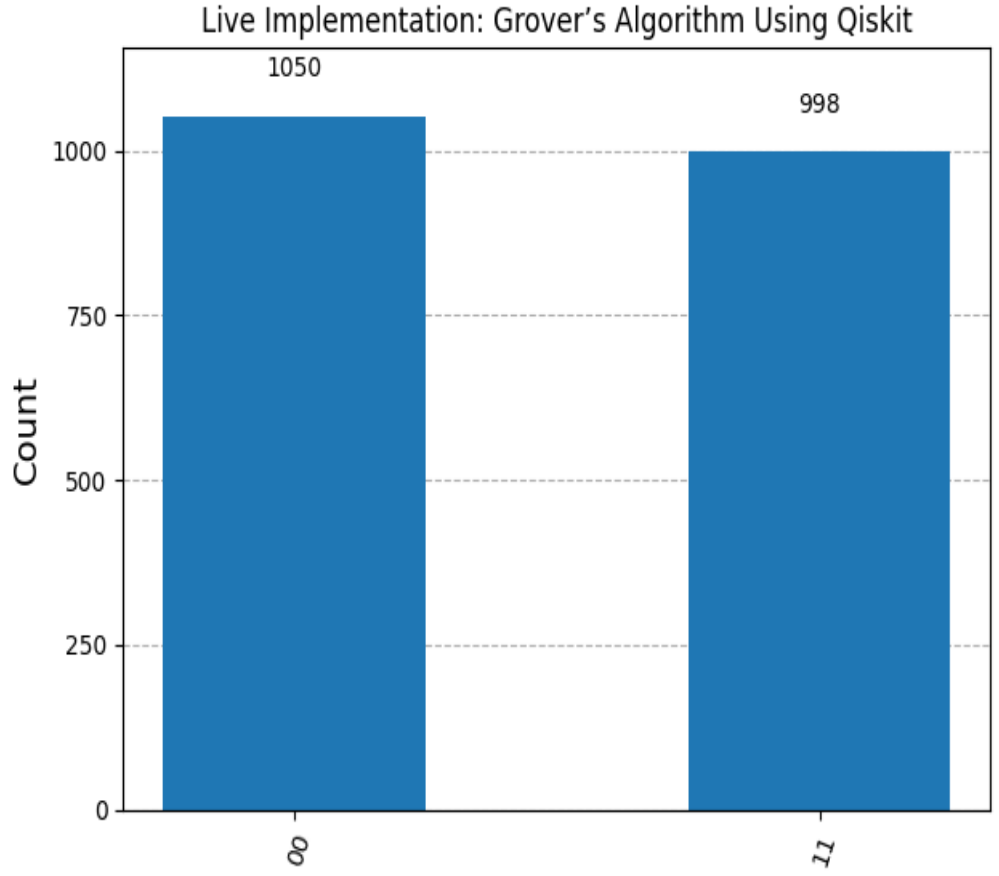
1 Iteration

Identifies the marked item with certainty for a two-qubit search.

Problem Context

Find one marked item among four possible items using amplitude amplification.


q_0
q_1
c
H
X
2
0
1


Live Implementation: Grover's Algorithm Using Qiskit
Count
1050
998
1000
750
500
250
0
00
11

Practical 6: Quantum Search for a Smart-Building Access Point

Scenario: Four candidate wireless access points. Target: AP-2 encoded as 10.

Candidate Access Points & Encoding

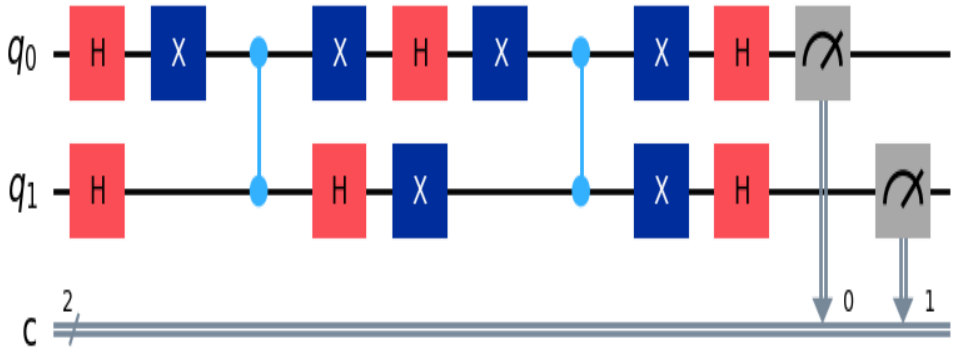
AP-0	00
AP-1	01
AP-2 (Target)	**10**
AP-3	11

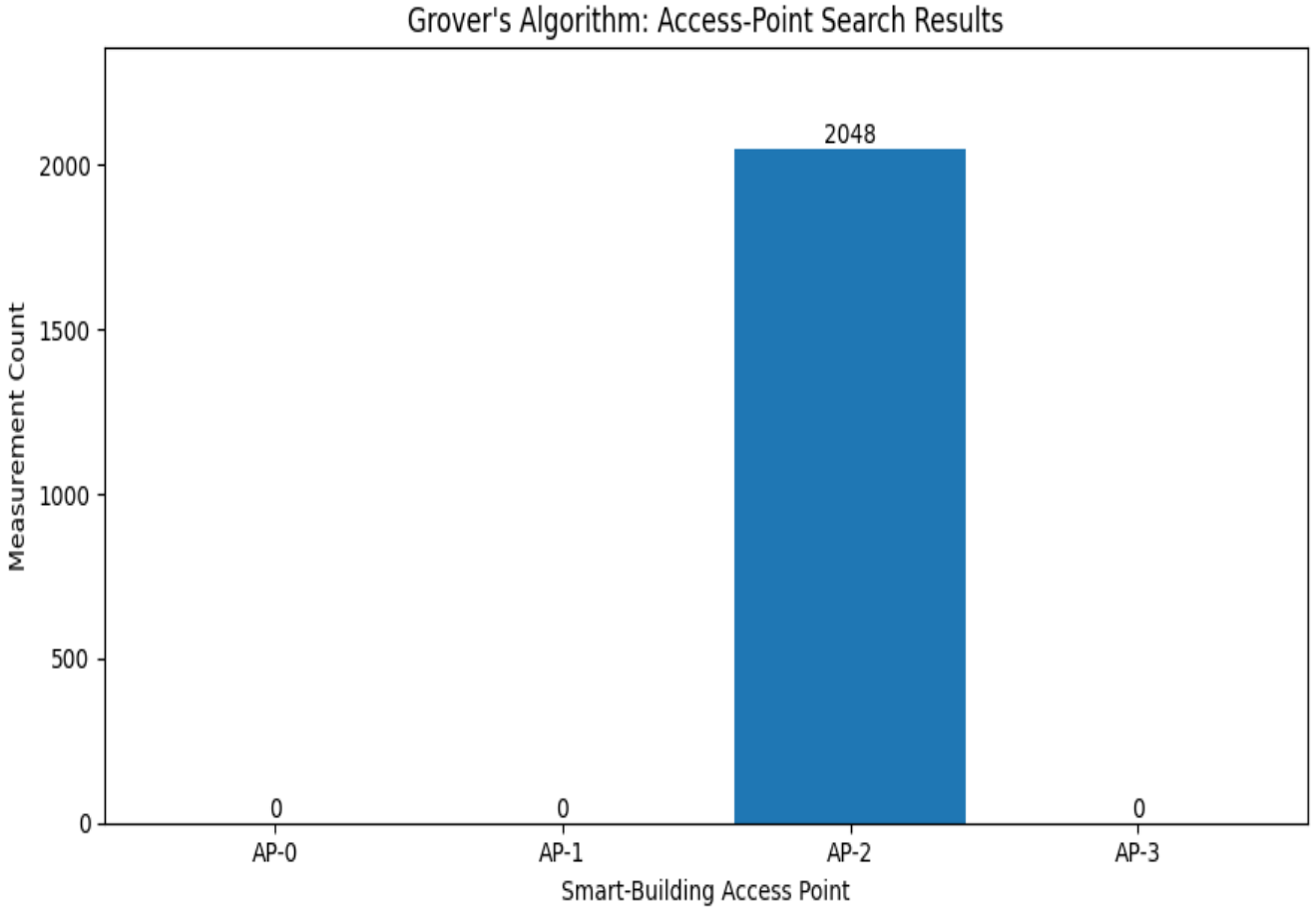
Quantum Workflow

Candidate set → State preparation → Target oracle → Diffusion → Measurement → Candidate identification

Important Distinction

This is a simplified algorithmic demonstration, not a complete Wi-Fi localization or access-point optimization system.


q_0
q_1
c
H
X
2
0
1


Grover's Algorithm: Access-Point Search Results
Measurement Count
0
500
1000
1500
2000
2048
0
0
0
AP-0
AP-1
AP-2
AP-3
Smart-Building Access Point

Practical 7: Simulate Quantum Noise

Objective: Compare ideal and noisy circuit execution

Noise Sources

- Gate errors
- Readout errors
- Decoherence via noise models

Experiment Steps

Run Bell-state circuit twice: (1) Ideal simulation, (2) Simulation with a noise model.

EVALUATION FOCUS

- Measurement distributions
- Frequency of unwanted outcomes
- Expected Bell-state outcome frequency

Compare Ideal and Noisy Results

Evaluation metrics and expected outcome analysis for noisy quantum simulation

Evaluation Metrics

1. Probability of the intended result
2. Frequency of unexpected outcomes
3. Deviation from the ideal distribution

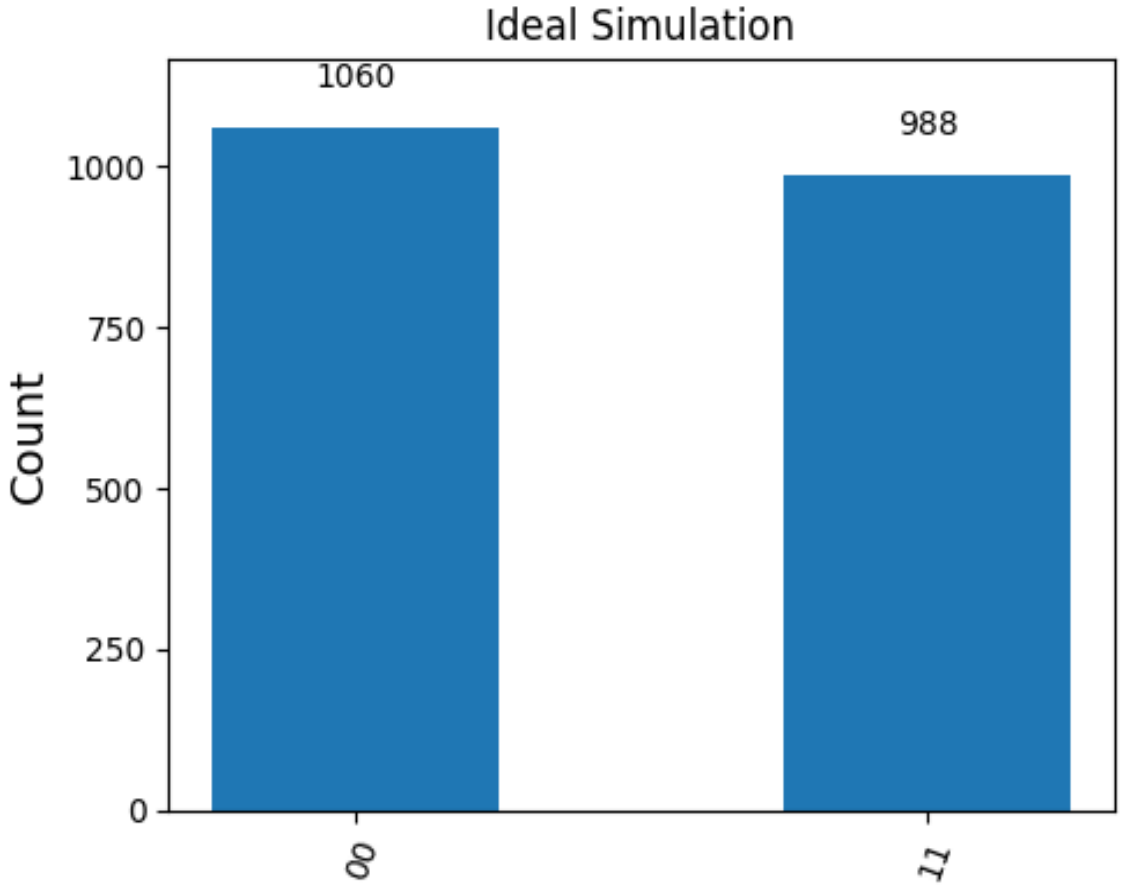
For the Bell State

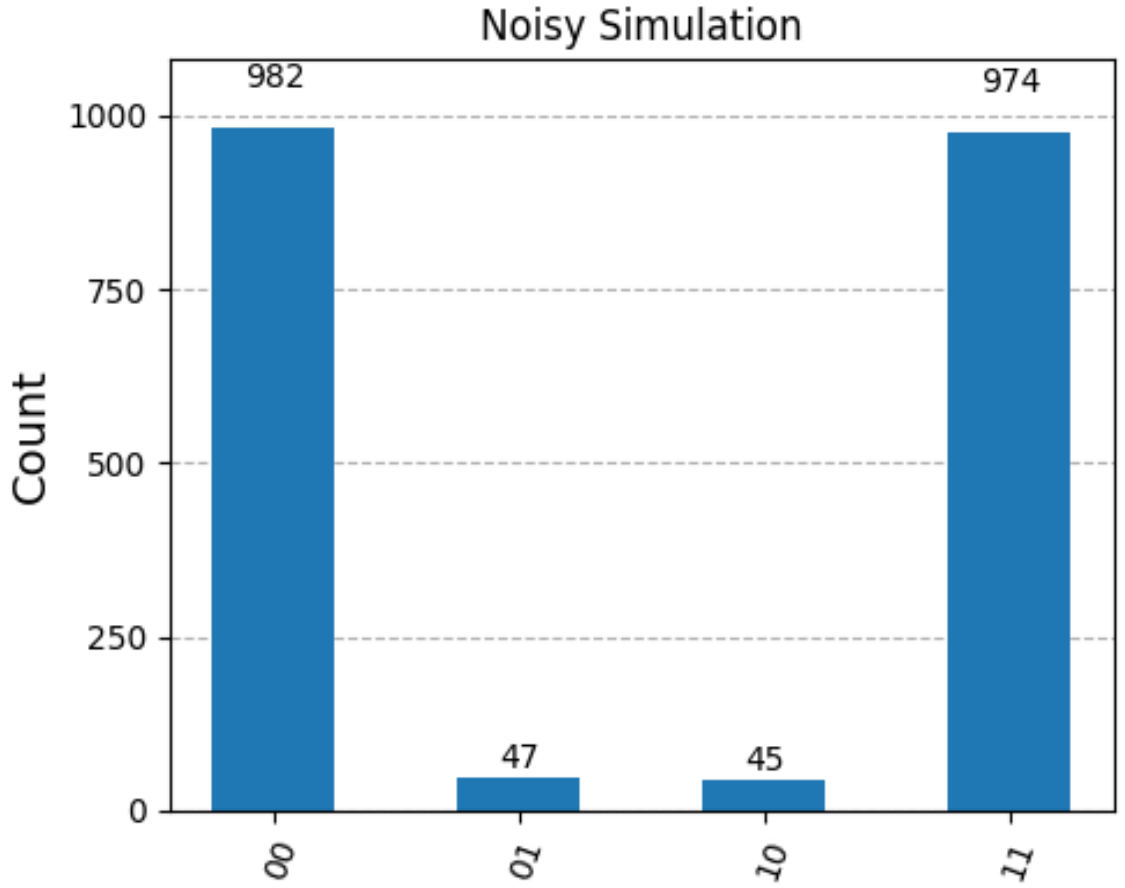
Expected outcomes: 00 and 11

Unexpected outcomes: 01 and 10

Key Question

How much does the noise model change the output distribution?


Ideal Simulation
Count
1060
988
1000
750
500
250
0
00
11


Noisy Simulation
Count
982
974
47
45
1000
750
500
250
0
00
01
10
11

Introducing QSimKit: My Quantum Simulation SDK

A Python-Based SDK for Quantum Circuit Construction and Simulation

Accessible Environment

Lowering entry barriers for students through intuitive APIs and clear mental models.

Internal Mechanics

Transparent state vector updates revealing full matrix and gate transformations.

Complete pipeline spanning circuit construction, simulation, and measurement.

Circuit Construction

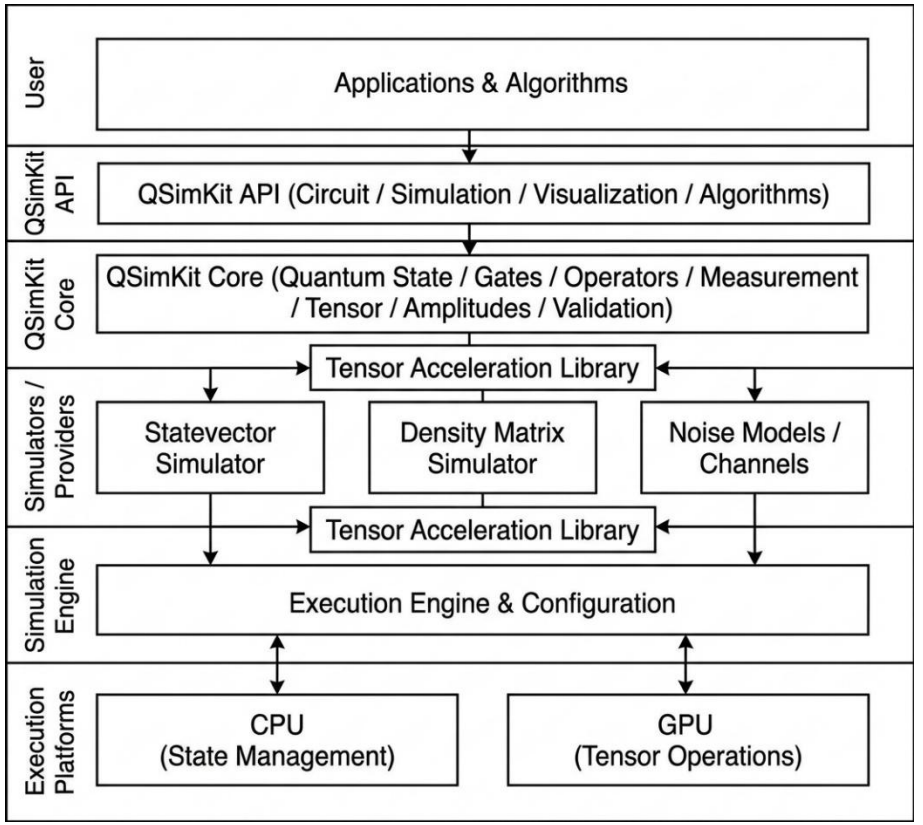
Define qubit registers, apply single-qubit and multi-qubit gates, and compose circuit routines.

Measurement Engine

Calculate theoretical probability distributions and sample shot-based measurement outcomes.

Represent exact complex quantum state vectors and compute state matrix operations.

Render text/graphical circuit diagrams and output histogram state visualizations.



Web Live Demonstration: QSimKit in Action

STEP 01

Construct Circuit

Initialize quantum circuit and apply Hadamard + CNOT gates to form a Bell state.

STEP 02

Execute Simulation

Pass circuit to QSimKit engine to apply matrix unitary transformations on state vectors.

STEP 03

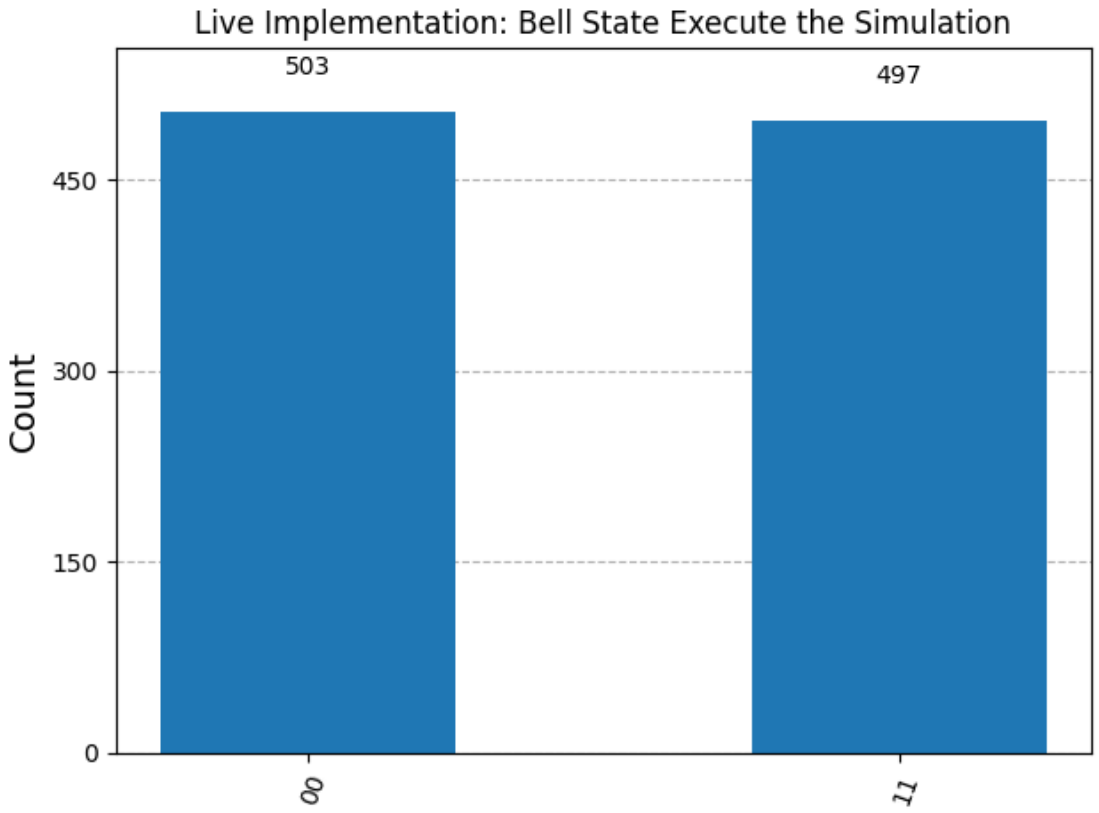
Display visual circuit diagram, state vector amplitudes, and measurement probabilities.

Expected Bell-State Behavior

TARGET STATE VECTOR

```
Circuit Diagram:
q0: ─H──●──M─
q1: ─────X──M─
```

Creates maximal two-qubit entanglement where measurement outcomes are strictly correlated.



Key Takeaways & System Architecture

